DATV-Easy VERSION 1.2x

Low-cost DATV broadcast from a PC with a LimeSDR mini or Adalm Pluto

Thanks for their great help with the development to all the beta testers.

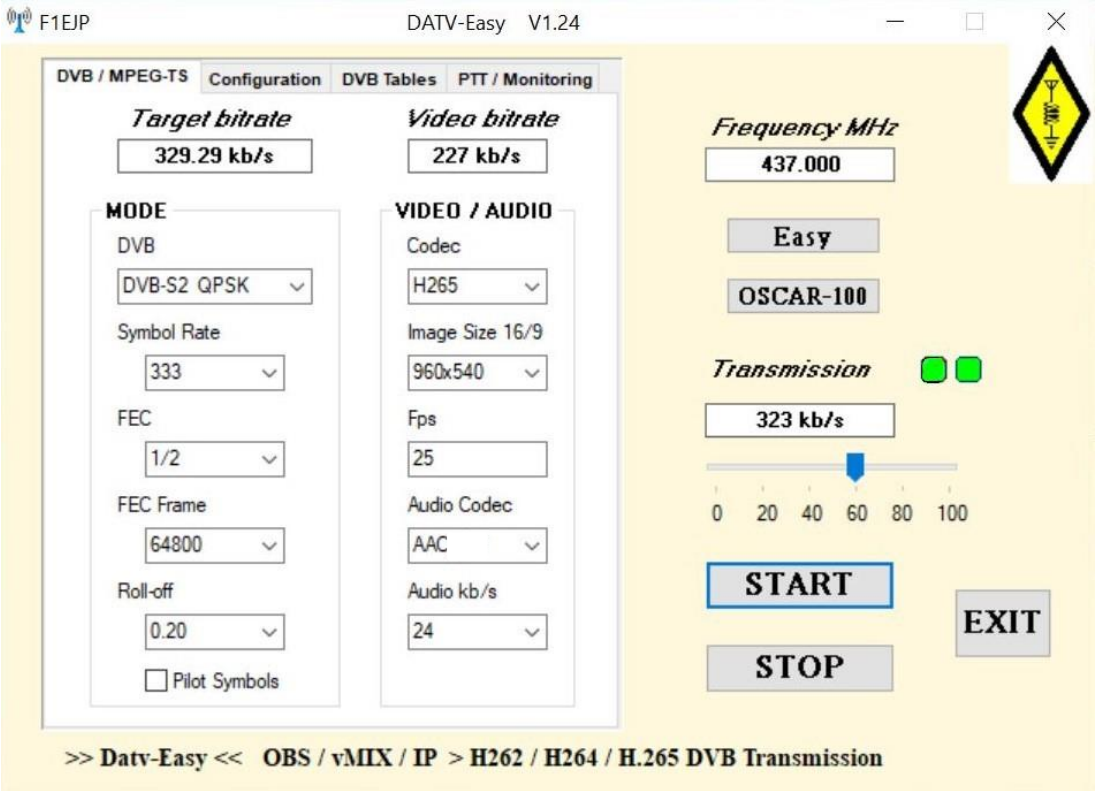
Prerequisites:

- A **LimeSDR mini SDR** with the latest Firmware (note the PortsDown dvb seems incompatible) or an **Adalm Pluto** with the manufacturer's original firmware. (compatible with Evariste firmware)
- An INTEL PC processor minimum Core I5, core I7 recommended or recent generation AMD.
- **Connection to USB 3 or Ethernet port.**
- If NVIDIA or AMD graphics card or Recent INTEL processor integrating GPU processor. Coding with hardware support is much more efficient than by soft
- If one of these prerequisites is not sufficient, you will have Drop Out in the program.

Functions :

- DATV-Easy allows transmission in DVB-S, DVB-S2 and DVB-T with a Limesdr mini or Adalm-Pluto and its original firmware with a Symbol Rate (SR) between 20 Ks/s and 2 Ms/ s
- Parameterization is facilitated for beginners by default preset values.
- You can choose your preferred frequencies of use "Easy Button" by editing the Frequency.csv file with Windows Notepad in C:\F1EJP (Values separated by ;)
- For Oscar 100 users you can choose the predefined frequencies (OSCAR-100 button)
- DATV-Easy uses FFMPEG with GPU processors from NVIDIA cards or recent INTEL processors to compress in H262, H264 and H265 -> Choose your encoder carefully.
- It can also code directly in soft but in this case the use is limited by the power of the processor above 333Ks or it can be saturated especially in H265.
- The Video and Audio source can be the free OBS software or vMIX or an already encoded iP stream.
- For audio you have the choice between MP2, AC3, AAC which allows lower bit rates.

Choose the SR (Symbol Rate or bandwidth) and FEC (error correction) and the software comes up with tested and roughly optimized defaults.

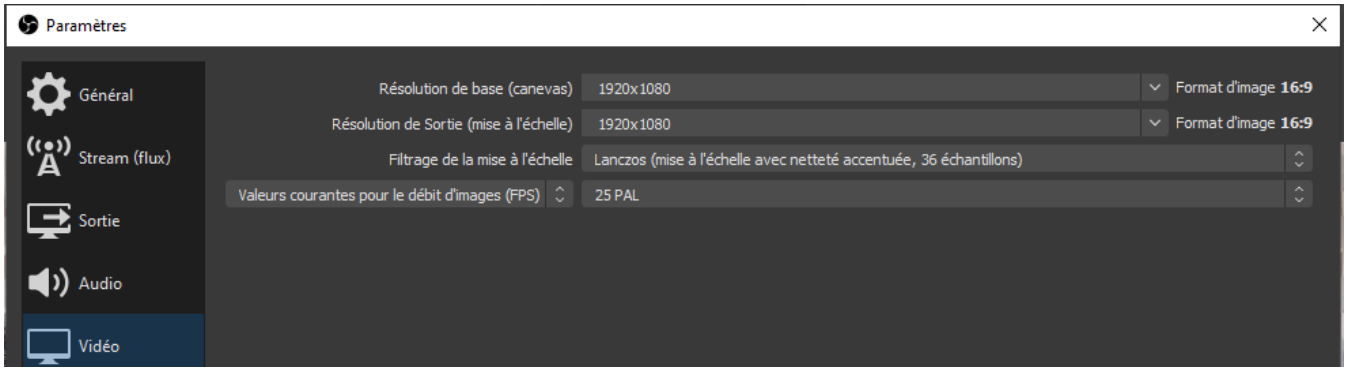


1/ If you haven't already, INSTALL OBS and the VirtualCam or vMIX plugin

- it is the best solution to stream videos from any source, camera, chart, film, etc.

https://obsproject.com/ https://www.vmix.com/

Many tutorials are available for their uses. With OBS here are just the settings to apply (will be grayed out with VirtualCam started)

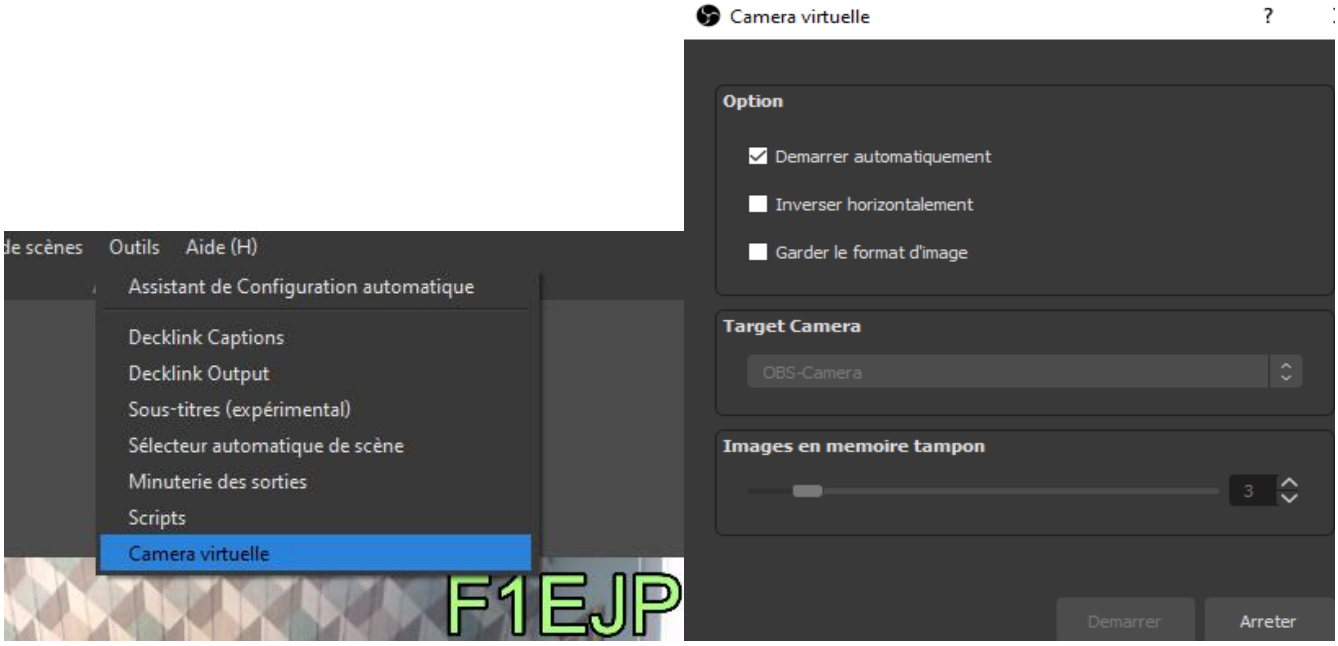


2/ For OBS Install VirtualCam

https://obsproject.com/forum/resources/obs-virtualcam.949/

Google is my friend for the latest versions and installation explanations.

On OBS, you no longer need to use the recording or streaming buttons, simply go to "Tools" then "Virtual Camera" and check "Start automatically"



3/ Installation and Configuration DATV-Easy

- **Install LIMESDR or PlutoSDR drivers**

LimeSDR-Mini driver installation - Myriad-RF Wiki (myriadrf.org)

https://wiki.analog.com/university/tools/pluto/drivers/windows

ATTENTION Adalm-Pluto can be used with the manufacturer's original firmware either via USB or with a USB / Ethernet converter. Several OMs have confirmed operation with F5OEO Firmware

Install

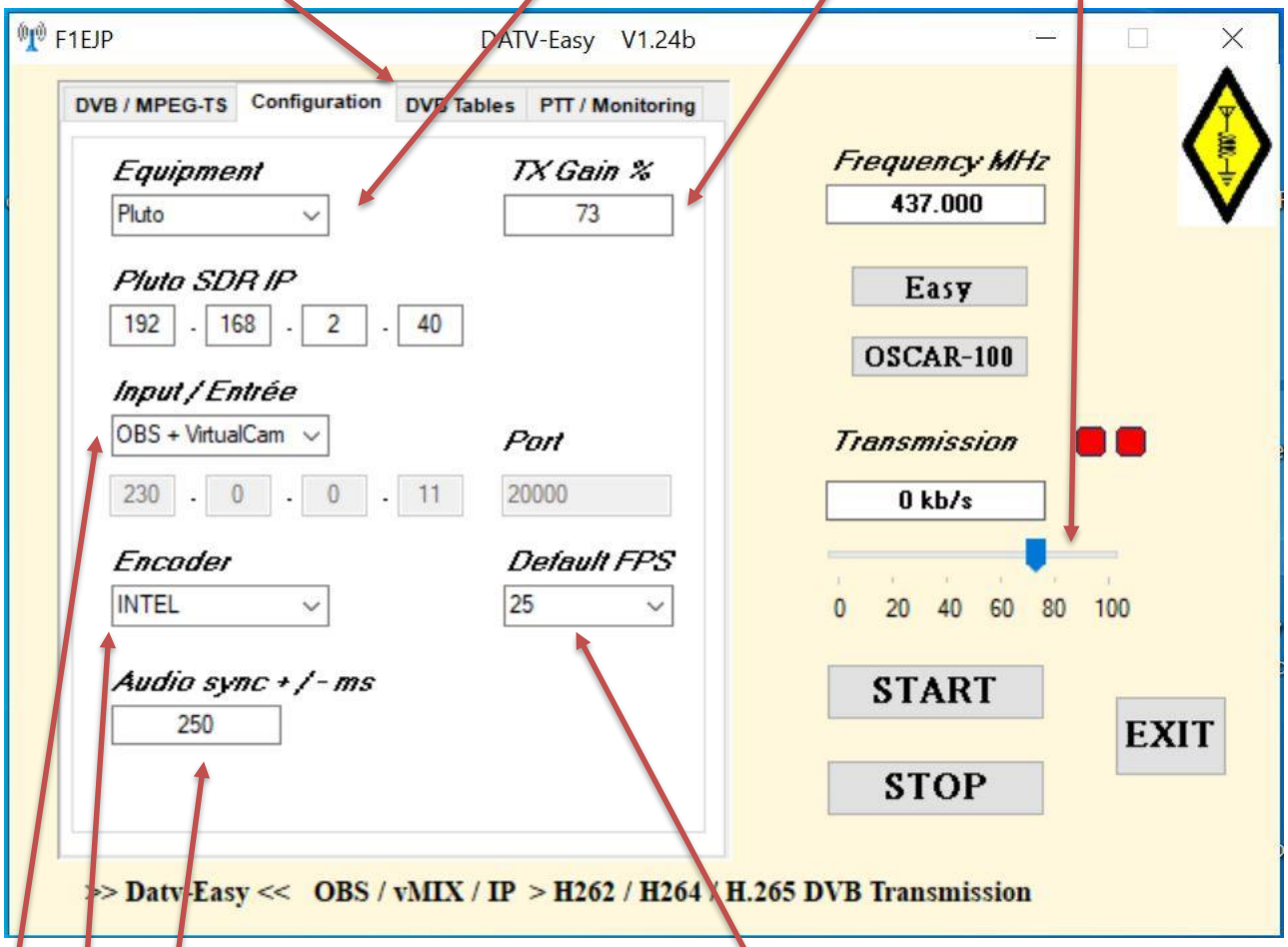
SetupF1EJP-DATV-Easy-V1.24.exe

Attention > During installation and on the first launch your antivirus can block it, you will have to accept if necessary because it is not a virus.

A Go to the "Configuration" tab»

Choice of SDR

Transmit Power



Audio / Video synchronization + / -

Default Frame rate at 25 or 30 frames / second

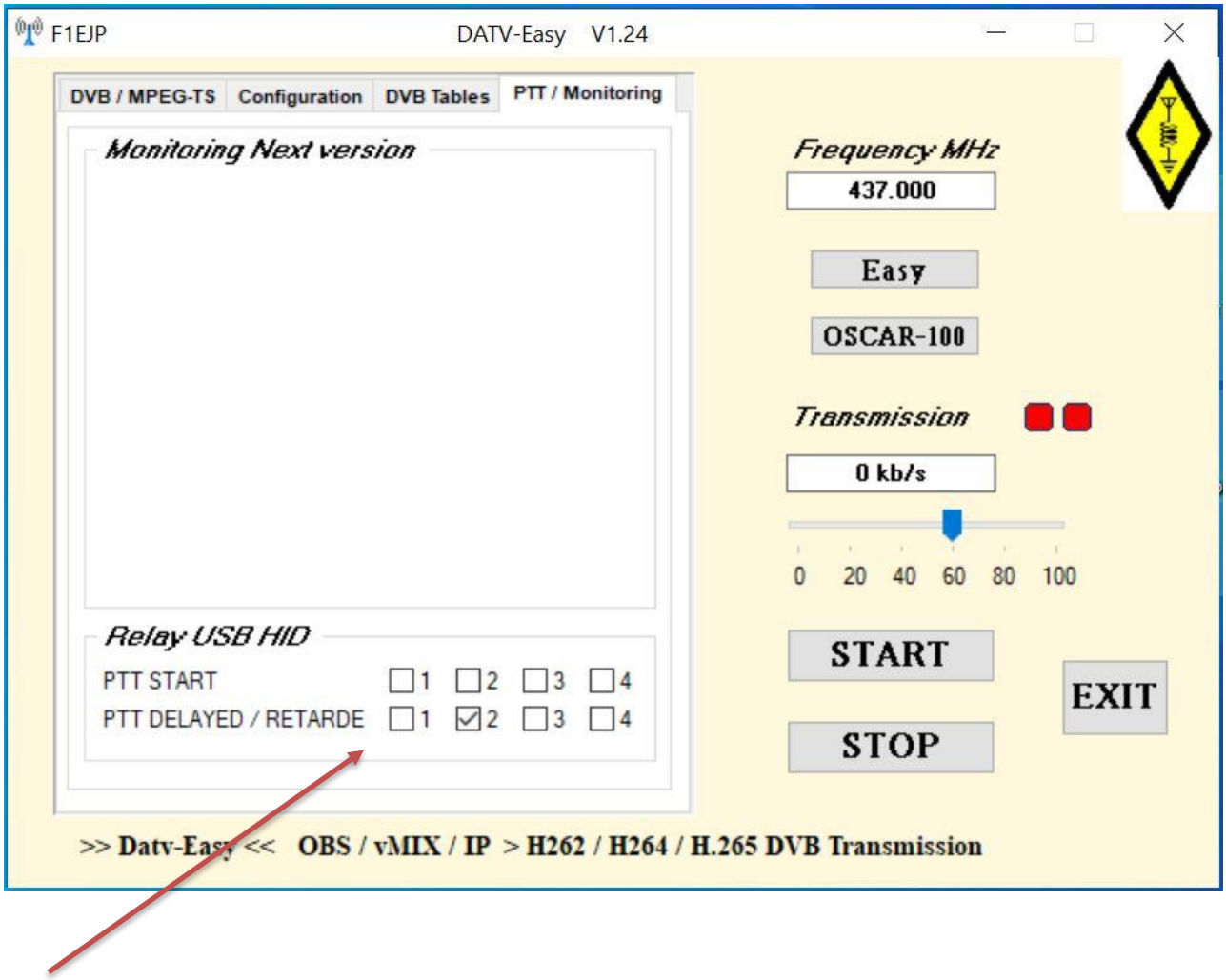
Choose support for encoding
(NVIDIA or AMD card or INTEL processor or software)

Choose the source: OBS or vMIX software or already coded IP stream (enter the input IP address and the port)

4/ PTT commands and monitoring

Monitoring by ETHERNET link for PTT control and monitoring of measured values remotely.

>> in development for next releases



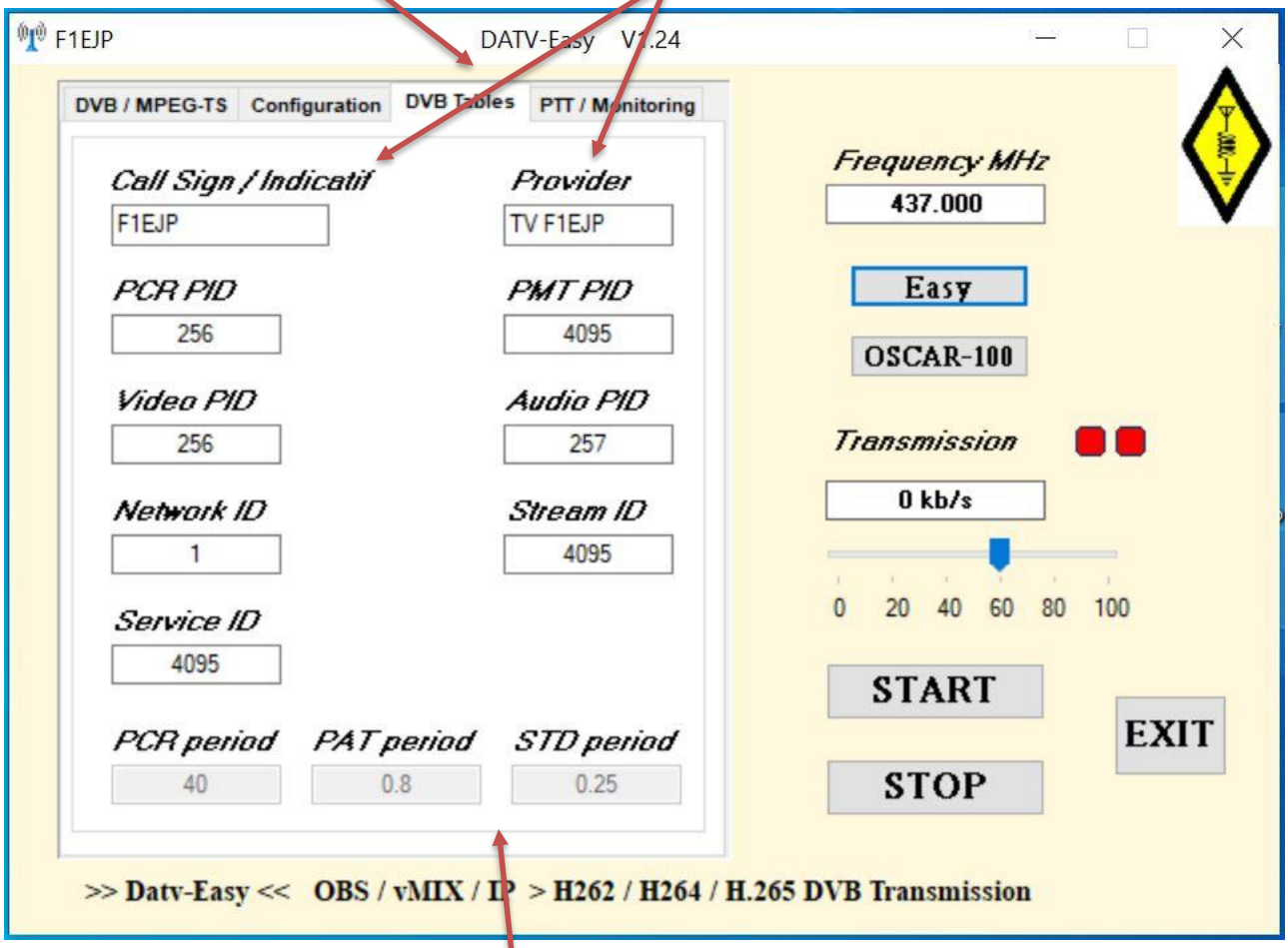
To control the power supplies of your amps without or with delay to avoid calibration peaks :

You can use USB HID relays without the need to install any driver.
Found on Ebay or Amazon for 7 to 17 €:

5/ Optional

Go to the “DVB Table” tab

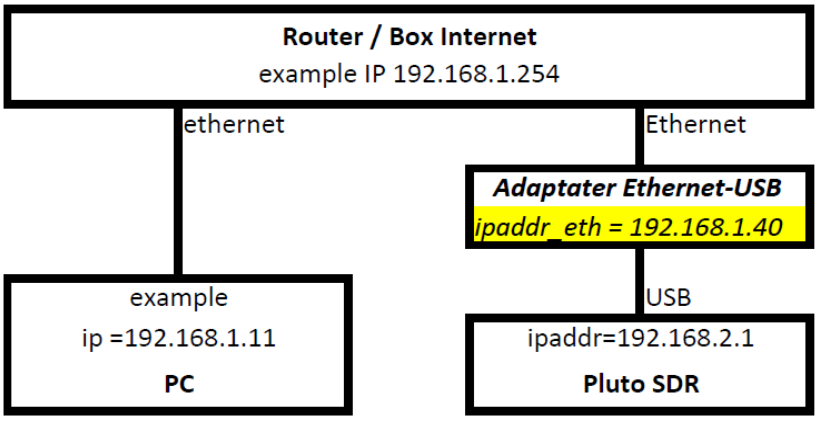
Fill in your callsign and provider



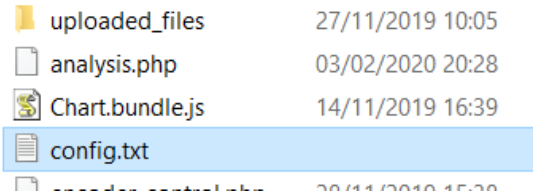
Optional:
If necessary, fill in the Identifications (PID, SID) of the different DVB streams, otherwise leave as default
Audio, Video etc. streams.
Attention the PCR PID takes the value of the video PID

For information: the periodic values of the tables according to the parameters chosen on the

6/ Connection Ethernet with Pluto and settings :



The PlutoSDR once connected as in the diagram, should automatically have an IP address given by your network router (router, internet box).
It is preferable to have a fixed IP address, which does not change.
So you have to modify the file which is in the root directory of the PlutoSDR player in the file *config.txt*



- Open the *config.txt*
- After the line *[USB_ETHERNET]*, modify the IP address by indicating a free IP address of your network. There are network analyzer software that allow you to list all the equipment connected to your network

The Pluto will have this fixed IP address on the next restart, in the example 192.168.1.40 to put in DATV-Easy > Pluto SDR IP

Ne pas modifier l'adresse section NETWORK mais celle de la section USB_ETHERNET.

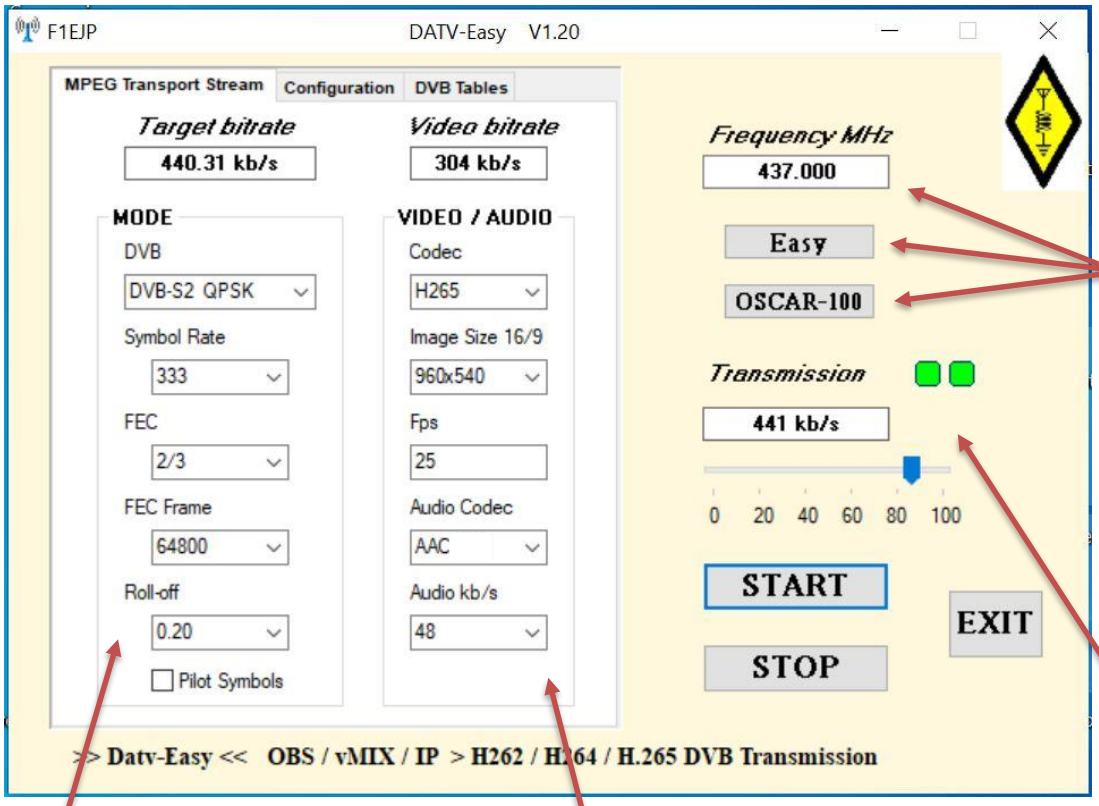
```
# Analog Devices PlutoSDR Rev.B (Z7010-AD9363)
# Device Configuration File
#   1. Open with an Editor
#   2. Edit this file
#   3. Save this file on the device USB drive
#   4. Eject the device USB Drive
# Doc: https://wiki.analog.com/university/tools/pluto/users/customizing

[NETWORK]
hostname = pluto
ipaddr = 192.168.2.1
ipaddr_host = 192.168.2.10
netmask = 255.255.255.0

[WLAN]
ssid_wlan =
pwd_wlan =
ipaddr_wlan =

[USB_ETHERNET]
ipaddr_eth = 192.168.1.40
netmask_eth = 255.255.255.0
gateway_eth = 192.168.0.254
```

7/ DATV-Easy use



Tap or choose your frequency with the buttons.

Modify C:\F1EJP\ Frequency .csv

First choose the type of DVB modulation

Choose Symbol Rate or Bandwidth and FEC Error Correction and or Guardfactor.

In DVBS2 the calculated bit rate also takes into account the parameters chosen in transmission for the FEC Frame and Pilots Symbol

In DVBT the calculated bit rate takes into account the bandwidth of the FEC and the GuardFactor

Choose the Video encoding: H264 or H265 or H262(mpeg2)

Choose Audio encoding: MP2, AC3, AAC

The image resolution, audio bit rate and frame rate are offered by default depending on the DVB parameters and the type of encoder used.

- You can modify them for testing. However if the settings are too high ffmpeg may no longer get the right bitrate and crashes.

The left LED displays the correct operation of the SDR and its driver

The light on the right is that of ffmpeg.

The display shows you the actual bitrate output from the encoder. It turns red if too high and 0 if there is a coding problem.

The slider allows you to adjust the power during transmission.

- Launch OBS or vMIX or the IP stream then click on **START** to start the broadcast (delay 12 seconds)
- **STOP** to stop
- **EXIT** to quit the software

Thank you for all your suggestions and test feedback. 73 Dominique F1EJP

I am regularly on the chat: https://eshail.batc.org.uk/wb/